ANILA MATHAI

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SKILLS & INTERESTS

- **Technical Skills:** Python, C, C++, Java, HTML, CSS, JavaScript, Data Structures and Algorithms, , AI/ML, Time series Forecasting, Object Detection, Google Colab, Streamlit, OOP, MySQL, DBMS Fundamentals
- **Soft Skills:** Leadership, Communication, Time Management, Critical Thinking, Team Player, Hardworking, Creativity, Decision- Making, Work Ethic, Problem Solving
- **Interests:** Machine Learning, Game Development, Cybersecurity

EDUCATION

- **Govt. Model Engineering College** 2026
  KTU, B.Tech in Computer Science Engineering *8.92*
- **Chavara Public School** 2021
  CBSE, 12th *96.2%*
- **Chavara Public School** 2019
  CBSE, 10th *96%*

PROJECTS

- **Auto-Architecture Explainer** ***Team Size: 4***
  AI/ML Developer *Ongoing*
  Technologies Used: *Python, Large Language Models (LLMs), Abstract Syntax Trees (AST), Clustering Algorithms, Graph Visualization*
  Developed a system to automatically analyze source code and generate software architecture diagrams with natural-language explanations. Implemented AST-based static code analysis and clustering to extract module dependencies and improve understanding of complex codebases.

- **Seatbelt Detector** ***Team Size: 1***
  AI/ML Developer *1 Month*
  Technologies used*: Python, PyTorch, MobileNetV2, OpenCV*
  Developed a lightweight image classification system to detect seatbelt usage using MobileNetV2, incorporating confidence-based filtering and a VLQ class to safely handle ambiguous or low-quality inputs.

- **Company Information Chatbot** ***Team Size: 1***
  Full Stack Developer *2 Months*
  Technologies Used*: Python, Streamlit, Gemini API, dotenv*
  Developed and deployed an energy advisory chatbot using Google's Gemini API to provide energy usage insights and answer queries based on company knowledge.

- **Arikomban** ***Team Size: 4***
  Frontend Developer *4 Months*
  Technologies used*: YOLOv8, OpenCV, PyTorch, Python, Telegram Bot API, MySQL, Roboflow*
  Contributed to the development of AriKomban, a real-time wild animal monitoring system leveraging YOLOv8 for detection and Telegram for instant alerts.

- **Blood Donation Management System** ***Team Size: 4***
  Frontend Developer *4 Months*
  Technologies Used: *HTML, CSS, JavaScript, Node.js, Express.js, MongoDB*
  Contributed to the development of a Blood Donation Management System that enables efficient donor matching and retrieval of contact details based on blood group compatability.

- **Candy Detection System** ***Team Size: 1***
  AI/ML Developer *1 Month*
  Technologies Used*: Python, Streamlit, YOLOv11s, OpenCV (cv2), Pillow (PIL), NumPy*
  Developed and deployed a deep learning-based quality control system using Ultralytics YOLOv11s and Streamlit to perform real-time visual inspection of candy products.

- **Store Sales Forecasting** ***Team Size: 1***
  AI/ML & Data Science Developer *2 Months*
  Technologies used*: Python, Pandas, Prophet, Matplotlib, Seaborn*
  Developed a time series forecasting model using Prophet on Kaggle's Store Sales dataset to predict future sales based on historical trends and external factors.

- **Ceramic Tile Defect Detection Model** ***Team Size: 1***
  AI/ML Developer *2 Months*
  Technologies used*: Python, Ultralytics YOLOv8, Roboflow*
  Developed and trained a cutting-edge YOLOv8 object detection model for ceramic tile defect detection.

COURSES & CERTIFICATIONS

- Pursuing an **Honours Degree** in **Machine Learning** offered by **KTU**.
- Obtained certification in **Deep Learning**, **NPTEL 2025**, conducted by **IIT Ropar**.
- Obtained certification in **Python Programming 2022** certified by **Techgyan**.
- Successfully completed a course on **CSS, Bootstrap,JavaScript, Web Development Course** recognized by **Udemy**.
- Obtained certification in **C++**, **Python**, **React.js**, **SQL Programming Essentials Bootcamp** certified by **LetsUpgrade** in collaboration with **NSDC National Skill Development Corporation**, **ITM Edutech Training Pvt**. **Ltd** and **GDG MAD**.

ACHIEVEMENTS & ACTIVITIES

- Attended **First International Conference on Trends in Engineering Systems and Technologies 2024**, organised by **Govt. Model Engineering College.**
- Took part in **Gen AI 2023**, a workshop organised by **Excel 2023**, the annual Techno-Managerial fest conducted by **Govt. Model Engineering College.**
- **Hobbies:** Photography, Portrait Sketching, Chess, Reading

REFERENCES

- **Prof. Dr. Mini M G**, Principal, Govt. Model Engineering College, Kochi, Email ID**:** **principal@mec.ac.in**
- **Prof. Dr. Binu V P**, HOD, Computer Science Engineering, Govt. Model Engineering College, Kochi, Email ID: **binuvp@mec.ac.in**